

Question

When sodium carbonate is added to a mineral containing element **A** and burned with oxygen, a sodium salt **B** containing **A** is obtained. After acidification and evaporative crystallization of **B**, colorless crystals **C** are formed. **C** readily dehydrates to yield a white solid **D**. When **D** reacts with SF_4 above 110°C , only gas **E** containing **A** with a mass fraction of 50.96% and another gas **F** are produced. **F** reacts with an aqueous solution of **C** to precipitate red crystals **G**. When **G** reacts with SbF_5 at -23°C in SO_2 in a 1:5 ratio, an ionic compound **H** and SbF_3 are formed in a 1:1 product ratio. Which of the following statements is correct:

- A.** The element **A** in **B** is in its highest oxidation state
- B.** In the infinite chain structure of **D**, there are three chemically distinct oxygen atoms.
- C.** The point group of molecule **E** is the same as that of CCl_4 .
- D.** The element **A** in **G** is in its lowest valence state
- E.** In the cation of **H**, the **A** atom has two coordination numbers.
- F.** There is a fourfold rotation axis for the anions in **H**

Answer

Correct Answer: **E**

Detailed Explanation

The key breakthrough point of the question lies in the gas **E**. **D** reacts with SF_4 to produce gas **E** and gas **F**. SF_4 is a commonly used fluorinating agent that can replace oxygen atoms in oxides with fluorine atoms. Therefore, **D** is an oxide of **A**, and **E** is a fluoride of **A**. The mass fraction of **A** in **E** is 50.96%, the atomic weight of **A** is M , and the chemical formula of **E** is **Double subscripts: use braces to clarify** (the atomic weight of F is 19). Then we have: $\frac{xM}{xM+19y} = 0.5096$. By trial and error, we can obtain the most suitable result: element **A** is Se, and **E** is SeF_4 .

CHECKPOINT

0.5 PTS

A is Se

CHECKPOINT

0.5 PTS

E is SeF_4

The point group of molecule **E** is C_{2v} . The two have the same point group. The point group of CCl_4 is T_d . The two are different. Option C is incorrect.

CHECKPOINT

0.5 PTS

Option C is incorrect

D is an oxide of Se, which reacts with SF_4 to produce SeF_4 , indicating that the valence state of Se in **D** is +4. So **D** is selenium dioxide (SeO_2). SeO_2 is a white solid, which matches the description. In the infinite chain structure of **D**, there are two types of oxygen atoms: terminal and bridging. Option B is incorrect.

CHECKPOINT

0.5 PTS

D is SeO_2

CHECKPOINT

0.5 PTS

Option B is incorrect

C is the hydrate of **D**, which easily loses water to obtain **D**. So **C** is selenous acid (H_2SeO_3). H_2SeO_3 is a colorless crystal, which matches the description.

CHECKPOINT

0.5 PTS

C is H_2SeO_3

B is acidified to obtain **C**. This indicates that **B** is a selenite. Although the first step is burning in oxygen, which is a strong oxidation process, selenite was not obtained due to the strong oxidizing property of hexavalent selenium.

B is Na_2SeO_3 , in which Se has a valence of +4, which is not the highest valence of +6. Option A is incorrect.

CHECKPOINT

0.5 PTS

B is Na_2SeO_3

CHECKPOINT

0.5 PTS

Option A is incorrect

SF_4 , as a commonly used fluorinating agent, can replace oxygen atoms in oxides with fluorine atoms and obtain SO_2 gas. **F** is SO_2 .

CHECKPOINT

0.5 PTS

F is SO_2

SO_2 has reducing properties and reacts with H_2SeO_3 to reduce it to red crystalline elemental Se. Analogous to the congener element S, **G** is Se_8 . The oxidation state of Se in it is 0, which is not the lowest oxidation state of -2. Option D is incorrect.

CHECKPOINT

0.5 PTS

G is Se_8

CHECKPOINT

0.5 PTS

Option D is incorrect

SbF_5 is a strong Lewis acid and a strong oxidizing agent. It will oxidize Se and form polymeric cations, commonly Se_4^{2+} , Se_8^{2+} . The corresponding anions may be SbF_6^- , $\text{Sb}_2\text{F}_{11}^-$. Combining the reaction ratio and the product ratio, we can obtain **H** as $\text{Se}_8(\text{Sb}_2\text{F}_{11})_2$. In it, Se_8^{2+} has a five-membered ring fused ring structure, and there are two coordination numbers, 2 and 3. Option E is correct. The $\text{Sb}-\text{F}-\text{Sb}$ bond angle of the anion $\text{Sb}_2\text{F}_{11}^-$ is not 180° . The anion does not have a four-fold rotation axis. Option F is incorrect.

CHECKPOINT

0.5 PTS

H is $\text{Se}_8(\text{Sb}_2\text{F}_{11})_2$

CHECKPOINT

0.5 PTS

Option E is correct

CHECKPOINT

0.5 PTS

Option F is incorrect